Topology and Geometry in Physics – Note 1

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Abstract

This is the first note on the course of topology and geometry in physics at the School of Fundamental Physics and Mathematical Sciences, Hangzhou Institute for Advanced Study, UCAS.

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1 Affine space

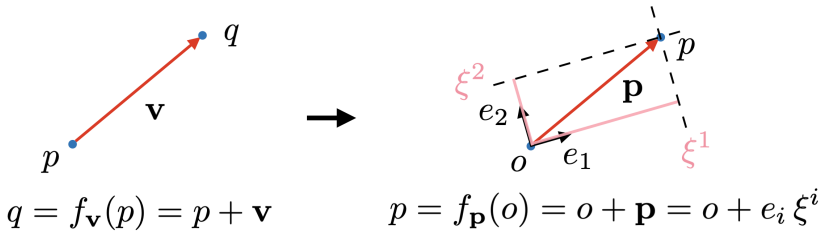

Fig. 1.1: Affine spaces (A, V) and (V, V) for $A = \mathbb{R}^2 = V$.

Given a point set A and a vector space V. For any point $a \in A$ and a vector $\mathbf{v} \in V$, we can get a new point $b \in A$ through the so-called *affine transformation*

$$\begin{aligned} f : A \times V &\longrightarrow A, \\ (a, \mathbf{v}) &\longmapsto b := f(a, \mathbf{v}) = a + \mathbf{v}, \end{aligned} \tag{1.1}$$

with such a structure we call a space an *affine space* denoted by a pair (A, V).

For every $a \in A$, we have an isomorphism from V to A:

$$\begin{aligned} f_a : V &\longrightarrow A, \\ \mathbf{v} &\longmapsto b = f_a(\mathbf{v}) = a + \mathbf{v}. \end{aligned} \tag{1.2}$$

Therefore, given a point $p \in A$ we can recognized the point p as a position vector $\mathbf{p} \in V = \mathbb{R}^n$ with respect to a reference point o as an *origin* in A by using the affine transformation $p = f_o(\mathbf{p}) = o + \mathbf{p}$. Let $\{e_i\}$ be a frame attached at o, the vector $\mathbf{p}$ can be written by $\mathbf{p} = e_i \, \xi^i$ with the coordinates $\{\xi^i\}$. Particularly $o = f_o(\mathbf{o}) = o + \mathbf{o}$ means that the point o can be represented by a zero vector $\mathbf{o} \in V$. Therefore A is isomorphic to a vector space V, and then we obtain an affine space (V, V) such that

$$\mathbf{p} = f_{\mathbf{o}}(\mathbf{p}) = \mathbf{o} + \mathbf{p} \quad \text{and} \quad \mathbf{q} = f_{\mathbf{p}}(\mathbf{v}) = \mathbf{p} + \mathbf{v} \tag{1.3}$$

with $\mathbf{v} \in V$, corresponding to

$$p = f_o(\mathbf{p}) = o + \mathbf{p} \quad \text{and} \quad q = f_p(\mathbf{v}) = p + \mathbf{v} \tag{1.4}$$

in (A, V), respectively.

In addition, for every $\mathbf{v} \in V$, we construct a map a as *translation* operation through a displacement vector $\mathbf{v}$ from the point a to point b:

$$\begin{aligned} f_{\mathbf{v}} : A &\longrightarrow A, \\ a &\longmapsto b = f_{\mathbf{v}}(a) = a + \mathbf{v}. \end{aligned} \tag{1.5}$$

For $A = \mathbb{R}^2 = V$, affine spaces (A, V) and (V, V) are illustrated by Fig. 1.1. With the structure of (V, V), there exist an infinitesimal displacement vector $\mathbf{v} = \mathbf{dp}$ from $\mathbf{p}$ to $\mathbf{q}$ such that $\mathbf{p}$ infinitesimally translate to $\mathbf{q} = \mathbf{p} + \mathbf{dp}$, which is shown in Fig. 1.2.

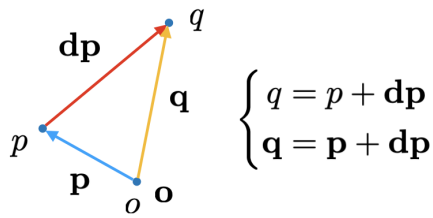

Fig. 1.2: Two points p and q joining by an infinitesimal displacement vector $\mathbf{dp}$.

2 Eclidean Vector Space

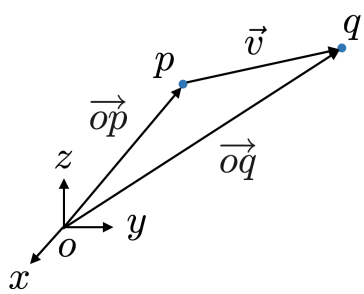

Fig. 2.1: A vector $\vec{v}$ in Euclidean space $\mathbb{R}^3$.

Vector space. In a n-dimensional Euclidean space $\mathbb{R}^n$, we always label a point by coordinates $\{x^i\} = \{x_1, x_2, \cdots, x_n\}$. We can define a point as the origin o with coordinates $(x_1, x_2, \cdots, x_n) = (0, 0, \cdots, 0)$. In addition, any point can be regarded as an end point of an arrow pointed from the origin o. We can naturally use the *Cartesian frame* $\{\vec{e}_1, \vec{e}_2, \cdots, \vec{e}_n\}$ as basis vector to describe such a directional arrow called *position vector*. Now we focus on the three-dimensional Euclidean space $\mathbb{R}^3$. Given two points p and q, shown in Fig. 2.1, and their coordinates are respectively labeled by (x, y, z) and (x', y', z'). We can have position vectors of points p and q written by $\overrightarrow{op} = x\,\vec{e}_x + y\,\vec{e}_y + z\,\vec{e}_z$ and $\overrightarrow{oq} = x'\,\vec{e}_x + y'\,\vec{e}_y + z'\,\vec{e}_z$ respectively.[†1] A *separation vector* $\vec{v}$ is always denoted by an arrow pointed from one point p to another point q. We assume that the separation vector between p and q is $\vec{v} := \overrightarrow{pq}$. The separation vector $\vec{v}$ is an arrow pointed from p to q, which can be regarded as a vector represents the action of the movement of a point from p to q. We always call this movement the *displacement vector* at point p. Then, from linear algebra, the vector $\vec{v}$ is the difference of $\overrightarrow{oq}$ and $\overrightarrow{op}$ and we have the relation

$$\begin{aligned} \vec{v} = \overrightarrow{pq} &= \underbrace{\overrightarrow{po}}_{-\overrightarrow{op}} + \overrightarrow{oq} = \overrightarrow{oq} - \overrightarrow{op} \\ &= (x' - x)\,\vec{e}_x + (y' - y)\,\vec{e}_y + (z' - z)\,\vec{e}_z \\ &\equiv v_x\,\vec{e}_x + v_y\,\vec{e}_y + v_z\,\vec{e}_z. \end{aligned} \tag{2.1}$$

[†1] We may sometimes recognize the points p and q by their position vectors denoted by $\mathbf{p} \equiv \overrightarrow{op}$ and $\mathbf{q} \equiv \overrightarrow{oq}$, respectively.

Notations of a vector space. Generally we prefer to write a n-dimensional vector by

$$\vec{v} = \sum_i v_i\,\vec{e}_i := v_i\,\vec{e}_i \overset{\text{or}}{=} \vec{e}_i\,v_i \quad (i = 1, 2, \ldots, n). \tag{2.2}$$

Here we use the so-called *Einstein's summation convention* in which the repeated index should be realized by the summation over the all components. Consider a three-dimensional vector $\vec{v}$, there are two notations for a vector can be presented here:

(i) We always use the components (or coordinates) with a *parentheses* or *square bracket* to represent the vector $\vec{v}$ denoted by a *row matrix* (or a *tuple*)

$$\mathsf{v} \equiv (v_i) = \begin{pmatrix} v_x & v_y & v_z \end{pmatrix} \overset{\text{or}}{=} (v_x, v_y, v_z) \quad \text{or} \quad \mathsf{v} \equiv [v_i] = \begin{bmatrix} v_x & v_y & v_z \end{bmatrix} \overset{\text{or}}{=} [v_x, v_y, v_z] \tag{2.3a}$$

or a *column matrix*

$$\mathsf{v} \equiv (v_i) = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \quad \text{or} \quad \mathsf{v} \equiv [v_i] = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}. \tag{2.3b}$$

We note that *matrices* are always denoted by letters in sans serif typeface. Conventionally, people always refer a vector to a column matrix, and such a matrix expression of the vector $\vec{v}$, cf. (2.1), is

$$\vec{v} \longrightarrow \mathsf{v} \equiv (v_i) = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = v_x\,\mathsf{e}_x + v_y\,\mathsf{e}_y + v_z\,\mathsf{e}_z \tag{2.4}$$

by recognizing basis vectors as the following basis matrices

$$\vec{e}_x \equiv \mathsf{e}_x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \vec{e}_y \equiv \mathsf{e}_y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{e}_z \equiv \mathsf{e}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.^{\dagger 2} \tag{2.5}$$

(ii) We always prefer the expression $\vec{v} = \vec{e}_i\,v_i$ (basis times component). The basis vectors are represented by a row matrix, while the corresponding components are represented by a column matrix. So a vector can easily correspond to the multiplication of a row matrix (basis matrix) and a column matrix (component matrix). Define basis and component matrices by

$$\vec{e}_i \longrightarrow \mathsf{e} = \begin{pmatrix} \vec{e}_1 & \vec{e}_2 & \cdots & \vec{e}_n \end{pmatrix}_{1\times n} \quad \text{and} \quad v_i \longrightarrow \mathsf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}_{n\times 1}. \tag{2.6}$$

Therefore, the vector $\vec{v}$ can be understood by the multiplication of matrices e and v:

$$\vec{v} = \vec{e}_i\,v_i \longrightarrow \vec{v} \equiv \mathsf{e}\cdot\mathsf{v}, \quad \text{or more precisely} \quad \vec{v} \equiv \mathsf{e}_{1\times n}\cdot\mathsf{v}_{n\times 1}. \tag{2.7}$$

[†2] This notation is always used as the notation of a *basis ket vector* in *quantum mechanics*.

Inner product. A *bilinear operator* g in a vector space V is introduced such that

$$g(a\,\vec{x}, b\,\vec{y}\,) = ab\,g(\vec{x}, \vec{y}\,), \tag{2.8}$$

where $\vec{x}$ and $\vec{y}$ are two vectors in V, and a and b are two real constants. We say that the inner product of two vectors is a bilinear operation of g which can be identified as a function of two *vectors*

$$g(\vec{v}, \vec{w}) = g(\vec{e}_i\,v_i, \vec{e}_j\,w_j) = v_i\,g(\vec{e}_i, \vec{e}_j)\,w_j. \tag{2.9}$$

We can define a matrix

$$\mathsf{G} = (g_{ij}), \tag{2.10}$$

where

$$g_{ij} \coloneqq g(\vec{e}_i, \vec{e}_j) = \vec{e}_i \cdot \vec{e}_j. \tag{2.11}$$

If $\{e_i\}$ is a *normalized* frame, then we would have

$$g_{ij} = \|\vec{e}_i\|\,\|\vec{e}_j\|\,\cos\theta_{ij} = \cos\theta_{ij}. \tag{2.12}$$

where θ_{ij} is the angle between $\vec{e}_i$ and $\vec{e}_j$. Consequently, g_{ij} given in a *normalized* frame is the *direction cosine* of the ith axis relative to the jth axis. Hence, the inner product can be expressed by a multiplication of matrix forms

$$g(\vec{v}, \vec{w}) = v_i\,g_{ij}\,w_j \equiv \mathsf{v}^{\mathrm{T}} \cdot \mathsf{G} \cdot \mathsf{w}, \tag{2.13}$$

where T stands for the *transpose* operation of a matrix and

$$\mathsf{v}^{\mathrm{T}} = \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix}, \quad \mathsf{G} = \begin{pmatrix} g_{11} & g_{12} & \cdots & g_{1n} \\ g_{21} & g_{22} & \cdots & g_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ g_{n1} & g_{n2} & \cdots & g_{nn} \end{pmatrix}, \quad \mathsf{w} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}. \tag{2.14}$$

If $\{\vec{e}_i\}$ is an *orthonormal* frame, then the matrix $g_{ij} = \delta_{ij}$, where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \longrightarrow \mathsf{G} = \mathrm{diag}\,(\underbrace{1, 1, \cdots, 1}_{n \text{ terms}}) = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix}_{n \times n} \tag{2.15}$$

is called the *Kronecker delta symbol*. In this situation, we can rewrite the inner product

$$g(\vec{v}, \vec{w}) = v_i\,\delta_{ij}\,w_j \equiv \mathsf{v}^{\mathrm{T}} \cdot \mathsf{G} \cdot \mathsf{w} \tag{2.16a}$$

$$= \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \tag{2.16b}$$

$$= \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \tag{2.16c}$$

$$= v_1 w_1 + v_2 w_2 + \cdots + v_n w_n, \tag{2.16d}$$

which is consist with the ordinary inner product of $\vec{v}$ and $\vec{w}$ in the Euclidean space, i.e., we obtain the identification of

$$\boxed{g(\vec{v}, \vec{w}) = v_i \, g_{ij} \, w_j = \vec{v} \cdot \vec{w},} \tag{2.17}$$

which leads to the norm of a vector $\vec{v}$ by

$$\boxed{\|\vec{v}\| = \sqrt{(v^1)^2 + (v^2)^2 + (v^3)^2}.} \tag{2.18}$$

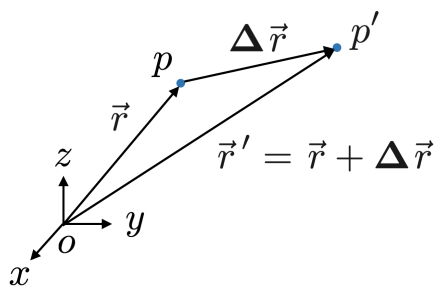

Fig. 2.2: A displacement vector $\boldsymbol{d}\vec{r}$ form the point p to the point q.

The line element. In $\mathbb{R}^3$, we naturally have the position vectors of points p and $p' \equiv q$ are respectively written as $\vec{r} = \vec{e}_x \, x + \vec{e}_y \, y + \vec{e}_z \, z$ and $\vec{r}\,' = \vec{r} + \boldsymbol{\Delta}\vec{r} = \vec{e}_x \, x' + \vec{e}_y \, y' + \vec{e}_z \, z'$, We note that $\boldsymbol{\Delta}\vec{r}$ is a displacement vector from p to q.[†3] This is shown in Fig. 2.2. To emphasize the *fixed* Cartesian frame, we sometime denote it by the notation of $\{\vec{e}_i\} = \{\vec{\delta}_i\}$ for $i = x, y, z$. Thus the displacement vector in *Cartesian coordinates* is usually read by

$$\boldsymbol{\Delta}\vec{r} = \overrightarrow{pq} = \overrightarrow{oq} - \overrightarrow{op} = \vec{r}\,' - \vec{r} \tag{2.19a}$$

$$= \vec{e}_x \, \Delta x + \vec{e}_y \, \Delta y + \vec{e}_z \, \Delta z \quad (\overset{\text{or}}{=} \vec{\delta}_i \, \Delta x^i). \tag{2.19b}$$

We note that $\boldsymbol{\Delta}$ in **bold-face** is used to denote the *variation of a vector* (a *vectorial* variation) in order to distinguish the ordinary variation of a function, which includes not only the difference of a function but also the addition law of vectors in linear algebra. The variation of a displacement vector is realized by

$$\begin{aligned} \boldsymbol{\Delta}\vec{r} &= \boldsymbol{\Delta}(\vec{e}_x \, x + \vec{e}_y \, y + \vec{e}_z \, z) \\ &= \underbrace{\boldsymbol{\Delta}\vec{e}_x}_{0} \, x + \vec{e}_x \, \Delta x + \underbrace{\boldsymbol{\Delta}\vec{e}_y}_{0} \, y + \vec{e}_y \, \Delta y + \underbrace{\boldsymbol{\Delta}\vec{e}_z}_{0} \, z + \vec{e}_z \, \Delta z. \end{aligned} \tag{2.20}$$

[†3] Generally we can denote the position vector of the point p by $\mathbf{p}$ (or by a matrix expression p) and write the displacement vector by $\boldsymbol{\Delta}\mathbf{p} \equiv \mathbf{p}' - \mathbf{p}$ (or by a matrix expression $\boldsymbol{\Delta}\mathsf{p}$) or simply $\boldsymbol{\Delta}p$. Correspondingly, its infinitesimal form should be $\boldsymbol{d}\mathbf{p}$ (or by a matrix expression $\boldsymbol{d}\mathsf{p}$) or simply $\boldsymbol{d}p$ respectively.

Therefore, due to the *invariant* basis of the Cartesian frame $\{\vec{e}_i\}$, the result obtained coincides with (2.19). The norm of the displacement vector $\boldsymbol{\Delta}\vec{r}$ is denoted by ΔS, which can be obtained by the inner product:

$$\Delta S := \|\boldsymbol{\Delta}\vec{r}\| = \sqrt{\boldsymbol{\Delta}\vec{r}\cdot\boldsymbol{\Delta}\vec{r}} = \sqrt{g(\boldsymbol{\Delta}\vec{r},\boldsymbol{\Delta}\vec{r})}. \tag{2.21}$$

We define the *line element* of a space as the square of the norm of $\boldsymbol{\Delta}\vec{r}$ by

$$\Delta S^2 = \boldsymbol{\Delta}\vec{r}\cdot\boldsymbol{\Delta}\vec{r} = g\big(\boldsymbol{\Delta}\vec{r},\boldsymbol{\Delta}\vec{r}\big) = g\big(\vec{e}_i\,\Delta x_i,\vec{e}_j\,\Delta x_j\big) = g\big(\vec{e}_i,\vec{e}_j\big)\Delta x_i\Delta x_j = g_{ij}\,\Delta x_i\Delta x_j. \tag{2.22}$$

In physics people are always interested in the *infinitesimal* form. If two points p and q are separated by an *infinitesimal distance* dS, then we have the *vectorial differential* of the position vector

$$\boldsymbol{\Delta}\vec{r} \quad\longrightarrow\quad \boldsymbol{d}\vec{r} = \overrightarrow{pq} = \overrightarrow{oq} - \overrightarrow{op} = \vec{r}\,' - \vec{r} \tag{2.23a}$$

$$= \vec{e}_x\,dx + \vec{e}_y\,dy + \vec{e}_z\,dz \quad (\overset{\text{or}}{=} \delta_i\,dx_i). \tag{2.23b}$$

So the *line element* of a space becomes

$$dS^2 = \boldsymbol{d}\vec{r}\cdot\boldsymbol{d}\vec{r} = g\big(\boldsymbol{d}\vec{r},\boldsymbol{d}\vec{r}\big) = g_{ij}\,dx_idx_j. \tag{2.24}$$

For two-dimensional Euclidean space, the displacement vector is $\boldsymbol{d}\vec{r} = \vec{e}_x\,dx + \vec{e}_y\,dy = \vec{e}_i\,dx_i$ and the corresponding line element with $g_{ij} = \delta_{ij}$ is

$$dS^2 = \delta_{ij}\,dx_idx_j \tag{2.25a}$$

$$\equiv \begin{pmatrix} dx & dy \end{pmatrix} \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix}. \tag{2.25b}$$

Similarly, the three-dimensional line element with $g_{ij} = \delta_{ij}$ and $\boldsymbol{d}\vec{r} = \vec{e}_x\,dx + \vec{e}_y\,dy + \vec{e}_z\,dz = \vec{e}_i\,dx_i$ is

$$dS^2 = \delta_{ij}\,dx_idx_j \tag{2.26a}$$

$$\equiv \begin{pmatrix} dx & dy & dz \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix}. \tag{2.26b}$$

Example. (Curvilinear Coordinates).

In stead of using the Cartesian coordinates, we can choose *curvilinear coordinates* for calculations. We would like to know the displacement vector $\boldsymbol{d}\vec{r} = \vec{r}\,' - \vec{r}$ and line element in these coordinate systems.

Polar coordinates in $\mathbb{R}^2$.

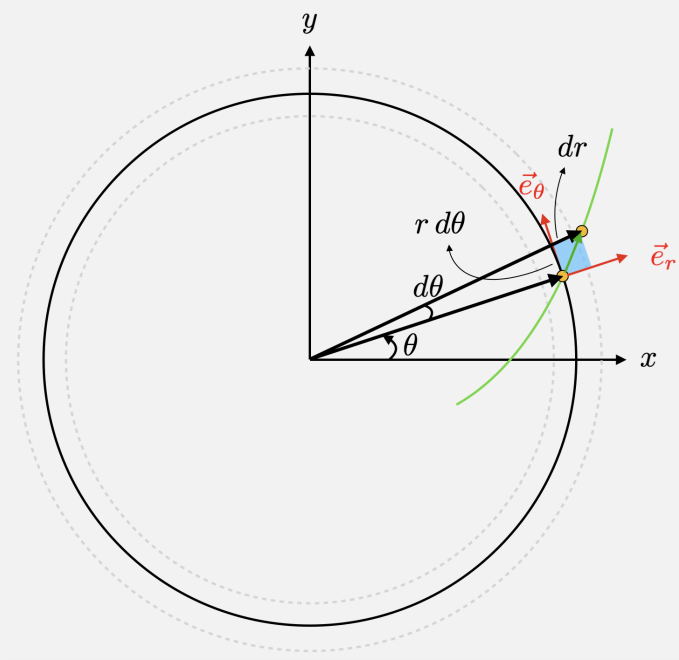

Fig. 2.3: Polar coordinates in $\mathbb{R}^2$.

Given a position vector

$$\vec{r}(x, y) = \vec{e}_x\, x + \vec{e}_y\, y, \tag{2.27}$$

where the coordinates (x, y) in terms of the polar coordinates are

$$\begin{cases} x = r\cos\theta, & (2.28a) \\ y = r\sin\theta. & (2.28b) \end{cases}$$

So we would like to consider the coordinate transformation

$$(x, y) \quad \longrightarrow \quad (r, \theta). \tag{2.29}$$

What about the coordinate transformation of the basis vector? That means we will have

$$\{\vec{e}_x, \vec{e}_y\} \quad \longrightarrow \quad \{\vec{e}_r, \vec{e}_\theta\}. \tag{2.30}$$

We know that basis vectors should be the tangent vectors along the corresponding direction of coordinates. Hence, we can obtain basis vectors through the derivatives with respect to the orresponding coordinates

$$\begin{aligned} \vec{V}_r = \frac{\partial \vec{r}}{\partial r} &= \frac{\partial(\vec{e}_x\, x + \vec{e}_y\, y)}{\partial r} \\ &= \underbrace{\frac{\partial \vec{e}_x}{\partial r}}_{0} r\cos\theta + \vec{e}_x \frac{\partial(r\cos\theta)}{\partial r} + \underbrace{\frac{\partial \vec{e}_y}{\partial r}}_{0} r\sin\theta + \vec{e}_y \frac{\partial(r\sin\theta)}{\partial r} \end{aligned}$$

$$= \vec{e}_x \cos\theta + \vec{e}_y \sin\theta \quad \longrightarrow \quad \boxed{\vec{e}_r := \frac{\vec{V}_r}{\|\vec{V}_r\|} = \vec{e}_x \cos\theta + \vec{e}_y \sin\theta,} \tag{2.31a}$$

$$\begin{aligned}
\vec{V}_\theta = \frac{\partial \vec{r}}{\partial \theta} &= \frac{\partial(\vec{e}_x\, x + \vec{e}_y\, y)}{\partial r} \\
&= \vec{e}_x \frac{\partial(r\cos\theta)}{\partial\theta} + \vec{e}_y \frac{\partial(r\sin\theta)}{\partial\theta} \\
&= -\vec{e}_x\, r\sin\theta + \vec{e}_y\, r\cos\theta \quad \longrightarrow \quad \boxed{\vec{e}_\theta := \frac{\vec{V}_\theta}{\|\vec{V}_\theta\|} = -\vec{e}_x \sin\theta + \vec{e}_y \cos\theta,}
\end{aligned} \tag{2.31b}$$

where the *scale factors* are defined as the *norm* of tangent vectors

$$\|\vec{V}_r\| = 1 := h_r, \tag{2.32a}$$

$$\|\vec{V}_\theta\| = r := h_\theta. \tag{2.32b}$$

The position vector in the polar coordinates is

$$\boxed{\begin{array}{ll} & \vec{r} = \vec{e}_x(r\cos\theta) + \vec{e}_y(r\sin\theta) = \vec{e}_r\, r \\ \Longrightarrow & \text{No cooordinate } \theta \text{ in the position vector.} \\ \Longrightarrow & \text{We would have a } \textit{symmetry} \text{ in the } \theta\text{-direction.} \end{array}} \tag{2.33}$$

By differentiating the position vector, the displacement vector can be regarded as a total differential of the vector $\vec{r}$:

$$\begin{aligned}
\boldsymbol{d}\vec{r} &= \boldsymbol{d}\big(\vec{e}_x\, x + \vec{e}_y\, y\big) \\
&= \underbrace{\boldsymbol{d}\vec{e}_x}_{0}(r\,\cos\theta) + \vec{e}_x\, \boldsymbol{d}(r\,\cos\theta) + \underbrace{\boldsymbol{d}\vec{e}_y}_{0}(r\,\sin\theta) + \vec{e}_y\, \boldsymbol{d}(r\,\sin\theta) \\
&= \vec{e}_x(dr\,\cos\theta - r\,\sin\theta\, d\theta) + \vec{e}_y(dr\,\sin\theta + r\,\cos\theta\, d\theta) \\
&= \underbrace{(\vec{e}_x\,\cos\theta + \vec{e}_y\,\sin\theta)}_{\vec{e}_r\, h_r = \vec{V}_r = \frac{\partial\vec{r}}{\partial r}}\, dr + \underbrace{(-\vec{e}_x\,\sin\theta + \vec{e}_y\,\cos\theta)\, r}_{\vec{e}_\theta\, h_\theta = \vec{V}_\theta = \frac{\partial\vec{r}}{\partial\theta}}\, d\theta.
\end{aligned} \tag{2.34}$$

On the other hand, the displacement vector can be directly realized in the polar coordinates by

$$\boldsymbol{d}\vec{r} = \boldsymbol{d}(\vec{e}_r\, r) = \boldsymbol{d}\vec{e}_r + \vec{e}_r\, dr. \tag{2.35}$$

By comparing (2.34) and (2.35), we have found that

$$\boxed{\boldsymbol{d}\vec{e}_r = \vec{e}_\theta\, d\theta.} \tag{2.36}$$

Alternatively, we can use (2.31) respectively to calculate $\boldsymbol{d}\vec{e}_r$ and $\boldsymbol{d}\vec{e}_\theta$:

$$\begin{cases} \boldsymbol{d}\vec{e}_r = -\vec{e}_x \sin\theta\, d\theta + \vec{e}_y \cos\theta\, d\theta = +\vec{e}_\theta\, d\theta, & (2.37a) \\ \boldsymbol{d}\vec{e}_\theta = -\vec{e}_x \cos\theta\, d\theta - \vec{e}_y \sin\theta\, d\theta = -\vec{e}_r\, d\theta. & (2.37b) \end{cases}$$

Hence, it results in that the displacement vector in the polar coordinates is given by

$$\boxed{\boldsymbol{d}\vec{r} = \frac{\partial \vec{r}}{\partial r}\, dr + \frac{\partial \vec{r}}{\partial \theta}\, d\theta = \vec{V}_r\, dr + \vec{V}_\theta\, d\theta \equiv \vec{e}_r(h_r\, dr) + \vec{e}_\theta(h_\theta\, d\theta),} \tag{2.38}$$

the sets $\{\vec{V}_r, \vec{V}_\theta\}$ and $\{\vec{e}_r, \vec{e}_\theta\}$ are both basis vector of the tangent space. The *line element* in coordinates (r, θ) is defined as the square of the norm of the displacement vector $\boldsymbol{d}\vec{r}$:

$$\begin{aligned} dS^2 = g(\boldsymbol{d}\vec{r}, \boldsymbol{d}\vec{r}) &= h_r^2\, dr^2 + h_\theta^2\, d\theta^2 \\ &\equiv \begin{pmatrix} h_r\, d\rho & h_\theta\, d\theta \end{pmatrix} \begin{pmatrix} h_r\, dr \\ h_\theta\, d\theta \end{pmatrix} \\ &= \begin{pmatrix} h_r\, d\rho & h_\theta\, d\theta \end{pmatrix} \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \begin{pmatrix} h_r\, dr \\ h_\theta\, d\theta \end{pmatrix} \\ &\overset{\text{or}}{=} \begin{pmatrix} d\rho & d\theta \end{pmatrix} \underbrace{\begin{pmatrix} h_r^2 & \\ & h_\theta^2 \end{pmatrix}}_{g_{ij}} \begin{pmatrix} dr \\ d\theta \end{pmatrix} \end{aligned} \tag{2.39}$$

with the metric tensor is

$$g_{ij} = \vec{V}_i \cdot \vec{V}_j = \begin{pmatrix} \vec{V}_r \cdot \vec{V}_r & \\ & \vec{V}_\theta \cdot \vec{V}_\theta \end{pmatrix} \equiv \begin{pmatrix} h_r^2 & \\ & h_\theta^2 \end{pmatrix} = \begin{pmatrix} 1 & \\ & r^2 \end{pmatrix} \quad (i, j = r, \theta). \tag{2.40}$$

In most situations, we would like to know the motion of a particle described, i.e., the trajectory of a particle, which is represented by the position vector $\vec{r}$. We needs to solve the equation of motion to and then obtain the solution $\vec{r}$. Therefore, it is hard to know the symmetry by reading the information from $\vec{r}$ initially, such as what we did in (2.33). However, the *displacement vector*(2.38) or the *line element* (2.39) contains most information of the geometric structure, we can start to calculate what we want, e.g., the equation of motion, we will explain in the following discussions, from one of them, and finally find the solution of $\vec{r}$.

Cylindrical coordinates in $\mathbb{R}^3$.

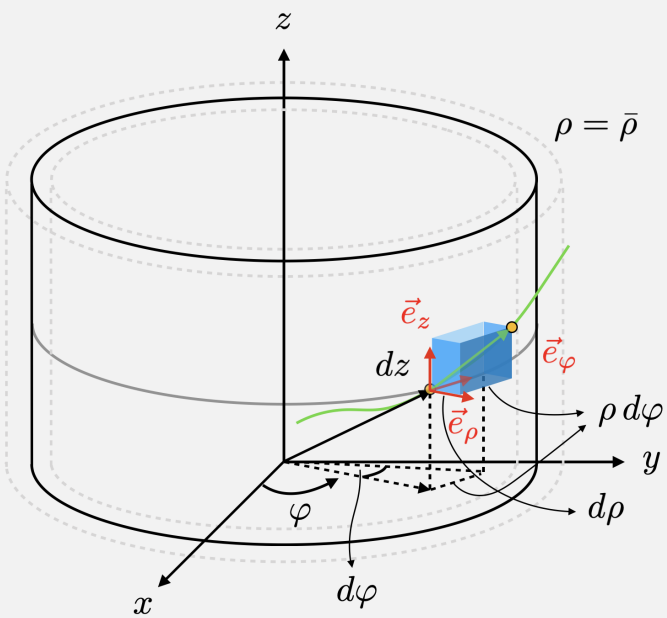

Fig. 2.4: Cylindrical coordinates in $\mathbb{R}^3$.

The coordinates (x, y, z) in terms of the cylindrical coordinates are

$$\begin{cases} x = \rho \cos\varphi, & (2.41a) \\ y = \rho \sin\varphi, & (2.41b) \\ z = z. & (2.41c) \end{cases}$$

Hence, the basis vectors should be

$$\begin{cases} \vec{e}_\rho = \vec{e}_x \cos\varphi + \vec{e}_y \sin\varphi, & (2.42a) \\ \vec{e}_\varphi = -\vec{e}_x \sin\varphi + \vec{e}_y \cos\varphi, & (2.42b) \\ \vec{e}_z = \vec{e}_z. & (2.42c) \end{cases}$$

The position vector is

$$\boxed{\begin{array}{ll} & \vec{r} = \vec{\rho} + \vec{z} = \vec{e}_\rho\, \rho + \vec{e}_z\, z \\ \Longrightarrow & \text{The } \varphi\text{-direction provides symmetry.} \end{array}}$$

The displacement vector

$$\begin{aligned} \boldsymbol{d}\vec{r} = \boldsymbol{d}(\vec{e}_\rho\, \rho + \vec{e}_z\, z) &= \underbrace{\boldsymbol{d}\vec{e}_\rho}_{\vec{e}_\varphi\, d\varphi}\, \rho + \vec{e}_\rho\, d\rho + \underbrace{\boldsymbol{d}\vec{e}_z}_{0}\, z + \vec{e}_z\, dz \\ &= \vec{e}_\rho\, d\rho + \vec{e}_\varphi(\rho\, d\varphi) + \vec{e}_z\, dz. \end{aligned} \tag{2.43}$$

The line element in (ρ, φ, z) can be written by

$$\begin{aligned} dS^2 &= d\rho^2 + \rho^2\, d\varphi^2 + dz^2 \\ &\equiv \begin{pmatrix} d\rho & d\varphi & dz \end{pmatrix} \underbrace{\begin{pmatrix} 1 & & \\ & \rho^2 & \\ & & 1 \end{pmatrix}}_{g_{ij}} \begin{pmatrix} d\rho \\ d\varphi \\ dz \end{pmatrix}. \end{aligned} \tag{2.44}$$

Spherical coordinates in $\mathbb{R}^3$.

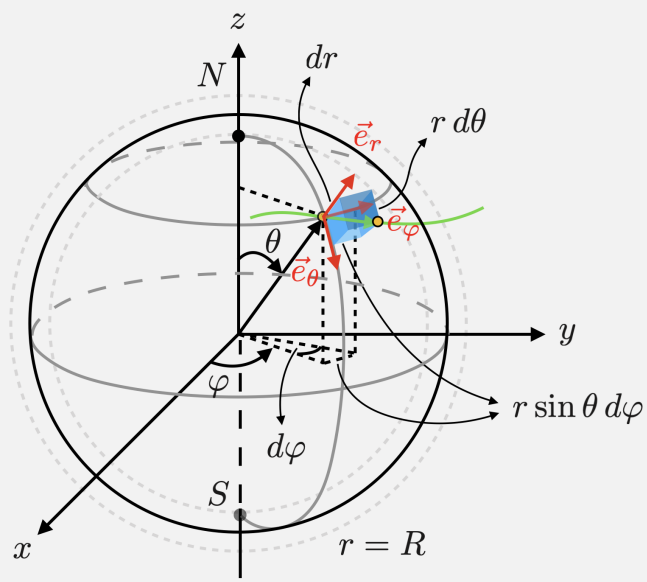

Fig. 2.5: Spherical coordinates in $\mathbb{R}^3$.

The coordinates (x, y, z) in terms of the spherical coordinates are

$$\begin{cases} x = r\sin\theta\cos\varphi, & (2.45a) \\ y = r\sin\theta\sin\varphi, & (2.45b) \\ z = r\cos\theta. & (2.45c) \end{cases}$$

Hence, the basis vectors should be

$$\begin{cases} \vec{e}_r = \vec{e}_x\sin\theta\cos\varphi + \vec{e}_y\sin\theta\sin\varphi + \vec{e}_z\cos\theta, & (2.46a) \\ \vec{e}_\theta = \vec{e}_x\cos\theta\cos\varphi + \vec{e}_y\cos\theta\sin\varphi - \vec{e}_z\sin\theta, & (2.46b) \\ \vec{e}_\varphi = -\vec{e}_x\sin\varphi + \vec{e}_y\cos\varphi. & (2.46c) \end{cases}$$

The position vector in the spherical coordinates is

$$\boxed{\begin{array}{ll} & \vec{r} = \vec{e}_r\, r \\ \Longrightarrow & \text{There exists a symmetry on the } (\theta, \varphi)\text{-surface.} \end{array}}$$

The displacement vector is

$$\boldsymbol{d}\vec{r} = \boldsymbol{d}(\vec{e}_r\, r) = \underbrace{\boldsymbol{d}\vec{e}_r}_{\vec{e}_\theta\, d\theta + \vec{e}_\varphi(\sin\theta\, d\varphi)}\, r + \vec{e}_r\, dr = \vec{e}_r\, dr + \vec{e}_\theta(r\, d\theta) + \vec{e}_\varphi(r\,\sin\theta\, d\varphi). \tag{2.47}$$

The line element in (r, θ, φ) can be written by

$$\begin{aligned} dS^2 &= dr^2 + r^2\, d\theta^2 + r^2\,\sin^2\theta\, d\varphi^2 \\ &\equiv \begin{pmatrix} dr & d\theta & d\varphi \end{pmatrix} \underbrace{\begin{pmatrix} 1 & & \\ & r^2 & \\ & & r^2\sin^2\theta \end{pmatrix}}_{g_{ij}} \begin{pmatrix} dr \\ d\theta \\ d\varphi \end{pmatrix}. \end{aligned} \tag{2.48}$$

3 Classical Differential Geometry - Curve Theory

3.1 Curves on $\mathbb{R}^2$

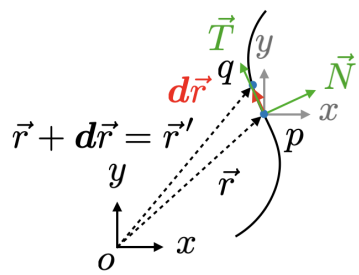

Fig. 3.1: A curve on $\mathbb{R}^2$.

Consider 2-dimensional Euclidean space, which is an $\mathbb{R}^2$ plane, which is constructed by a *fixed* frame. So any point in this plane can be described by a set coordinate function with respect to a reference point o, a 2-dimensional *Cartesian coordinate system*. A curve can be realized by a time series set of points, and hence, we parameterize a curve as a vector-valued function of one variable $t \in \mathbb{R}$, i.e., $\vec{r}(t) = \big(x(t), y(t)\big)^T$. The *velocity* vector at point $\vec{r}$ is given by

$$\vec{v}(t) = \dot{\vec{r}}(t) := \frac{\boldsymbol{d}\vec{r}}{dt} = e_x \frac{dx}{dt} + e_y \frac{dy}{dt} = \big(\dot{x}(t), \dot{y}(t)\big)^T, \tag{3.1}$$

where $\boldsymbol{d}\vec{r} = e_x\, dx + e_y\, dy + e_z\, dz$ and the *norm* of $\vec{v}$ is

$$\|\vec{v}\,\| = \|\dot{\vec{r}}(t)\| = \sqrt{\dot{\vec{r}} \cdot \dot{\vec{r}}} = \sqrt{\dot{x}^2 + \dot{y}^2}\,, \tag{3.2}$$

where $\dot{x} := dx/dt$. Then the arc length between a interval $[a, t]$ can be regarded as a function of parameter t given by

$$s = \int_a^t \|\dot{\vec{r}}(t')\|\, dt' \equiv f(t). \tag{3.3}$$

It can be shown that by using s as a parameter of a curve $\vec{r}(t(s))$ with $t = f^{-1}(s)$, the norm of the velocity is normalized, i.e.,

$$\|\vec{r}\,'(s)\| = \sqrt{x'^2 + y'^2} = \sqrt{\dot{x}^2 + \dot{y}^2}\,\frac{dt}{ds} = \|\dot{\vec{r}}\,\|\,\frac{dt}{ds} = \frac{df}{dt}\frac{dt}{ds} = 1, \tag{3.4}$$

where $x' := dx/ds$. Therefore, we can define a *unit tangent vector* as a *velocity* vector parameterized by s

$$\boxed{\vec{T} := \vec{r}\,'(s).} \tag{3.5}$$

By differentiate $\vec{r}\,' \cdot \vec{r}\,' = 1$ with respect to s again, we obtain

$$\vec{r}\,'' \cdot \vec{r}\,' + \vec{r}\,' \cdot \vec{r}\,'' = 0 \quad \Longrightarrow \quad \vec{r}\,'' \cdot \vec{r}\,' = 0, \tag{3.6}$$

so the *acceleration* vector $\vec{a}(s) := \vec{r}\,'' = \vec{T}\,'$ is normal to $\vec{T} = \vec{r}\,'$. Due to the orthonormality, we can define a *principal normal vector*

$$\boxed{\vec{N} = \frac{\vec{r}\,''}{\|\vec{r}\,''\|},} \tag{3.7}$$

which is a *unit* normal vector. We can rewrite (3.7) as

$$\boxed{\vec{T}\,' = \kappa(s)\,\vec{N}} \tag{3.8}$$

related to the normal vector $\vec{N}$, which spans a space normal to the tangent space spanned by $\vec{T}$ at any point of the curve $\vec{r}(s)$. The factor

$$\boxed{\kappa(s) := \|\vec{r}\,''\| = \|\vec{a}(s)\|} > 0 \tag{3.9}$$

as the *magnitude* of the acceleration vector is defined as the *curvature* of the curve $\vec{r}(s)$ at s.

Kinematically, the curvature of a space can be realized by the motion of a particle:

$$\text{Curvature} \sim \text{Acceleration}. \tag{3.10}$$

Now we can use $\vec{T}$ and $\vec{N}$ as a frame $(\vec{r}; \vec{T}, \vec{N})$ to span a two-dimensional space which is called the *osculating plane*.

Because of $\vec{T} \cdot \vec{N} = 0$, we can differentiate the relation and obtain a new equation

$$0 = \vec{T}\,' \cdot \vec{N} + \vec{T} \cdot \vec{N}\,' = \kappa \vec{N} \cdot \vec{N} + \vec{T} \cdot \vec{N}\,' \quad \Longrightarrow \quad \vec{T} \cdot \vec{N}\,' = -\kappa \quad \text{or} \quad \boxed{\vec{N}\,' = -\kappa \vec{T},} \tag{3.11}$$

where $\vec{N}\,'$ can be uniquely determined by $\vec{T}$ due to $\vec{N}\,' \cdot \vec{N} = 0$. As a result, we have the derivation equations of a frame $(\vec{r}; \vec{T}, \vec{N})$:

$$\begin{cases} \vec{r}\,' = \quad +\vec{T}, \\ \vec{T}\,' = \qquad\quad +\kappa \vec{N}, \\ \vec{N}\,' = -\kappa \vec{T}, \end{cases} \quad \Longrightarrow \quad \begin{pmatrix} \vec{r}\,' \\ \vec{T}\,' \\ \vec{N}\,' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \kappa \\ -\kappa & 0 \end{pmatrix} \begin{pmatrix} \vec{T} \\ \vec{N} \end{pmatrix}, \tag{3.12}$$

which is called the *Frenet-Serret formula* in $\mathbb{R}^2$.

Gauss map. Gauss map G of a curve is a mapping which globally send all the points p of a curve to an *unit circle* S^1 (a Gauss circle) centered at C and send the corresponding normal vector $\vec{N}$ to a radius vector from C pointing to S^1, which is shown as Fig. 3.2. Therefore, $\vec{N}$ can be represented as a position vector of a point on S^1.

It is interesting to find a correspondence between the curve γ and the Gauss circle S^1:

$$\vec{r} \text{ of a curve } \gamma \ \sim \ \vec{N} \text{ of } S^1 \quad \Longrightarrow \quad \boldsymbol{d}\vec{r} \text{ of a curve } \gamma \ \sim \ \boldsymbol{d}\vec{N} \text{ of } S^1. \tag{3.13}$$

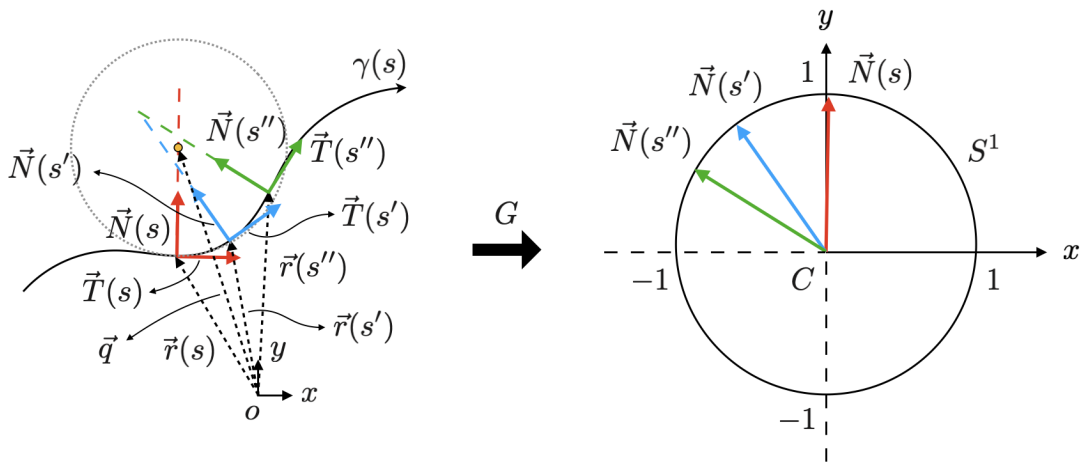

Fig. 3.2: The Gauss map G of a curve.

Let's consider two normal vectors $\vec{N}(s)$ and $\vec{N}(s')$ with respect to two infinitesimally nearby points $\vec{r}(s)$ and $\vec{r}(s')$, where $s' = s + ds$ is infinitesimally close to s. We can expand $\vec{N}(s')$ at s:

$$\begin{aligned}\vec{N}(s') &= \vec{N}(s+ds)\\ &\approx \vec{N}(s) + \vec{N}'(s)ds\\ &= \vec{N}(s) + \big(-\kappa(s)\vec{T}(s)\big)ds\\ &= \vec{N}(s) + \big(-\kappa(s)\,ds\big)\vec{T}(s)\,,\end{aligned} \tag{3.14}$$

which is the parametrization of a point under the Gauss map. Thus, we know the distance between two infinitesimal points represented by $\vec{N}(s)$ and $\vec{N}(s')$ on Gauss circle given by

$$\|\vec{N}(s') - \vec{N}(s)\| = \|\boldsymbol{d}\vec{N}\| = \kappa(s)\,ds\,. \tag{3.15}$$

And we also have the displacement vector

$$\boldsymbol{d}\vec{r} \equiv \vec{r}(s') - \vec{r}(s) \approx \big(\cancel{\vec{r}(s)} + \vec{r}\,'(s)\,ds\big) - \cancel{\vec{r}(s)} = \vec{T}(s)\,ds \tag{3.16}$$

with the norm of

$$\|\boldsymbol{d}\vec{r}\| = \|\vec{r}(s') - \vec{r}(s)\| \approx ds\,. \tag{3.17}$$

Therefore, in the *local* region, the ratio of the length between two points on the Gauss circle and curve, i.e., $\|\boldsymbol{d}\vec{N}\|/\|\boldsymbol{d}\vec{r}\|$ can be calculate by

$$\frac{\|\vec{N}(s') - \vec{N}(s)\|}{\|\vec{r}(s') - \vec{r}(s)\|} = \frac{\|\boldsymbol{d}\vec{N}\|}{\|\boldsymbol{d}\vec{r}\|} \approx \frac{\kappa(s)\,ds}{ds} = \kappa(s)\,, \tag{3.18}$$

which measure the curvature of a curve, $\kappa(s)$, at the neighborhood of a local point $\vec{r}$.

According to the example of a circle, the curvature is $\kappa = 1/r$. So, we can assume that the position vector of the center of osculating circle is identified by the the vector

$$\vec{q} = \vec{r} + r\vec{N} = \vec{r} + (1/\kappa)\vec{N}. \tag{3.19}$$

Taking the derivative of $\vec{q}$ with respect to s, we obtain

$$\vec{q}\,' = \vec{r}\,' + \frac{1}{\kappa}\vec{N}' = \vec{T} + \frac{1}{\kappa}(-\kappa\vec{T}) = 0\,, \tag{3.20}$$

which means that $\vec{q}$ is fixed, i.e., $\vec{q}$ points to the center of the osculating circle with radius $1/\kappa$. By considering the Gauss map of a circle. The radius vector of the the Gauss circle should be $\vec{N}$, and its center C corresponds to the center of the osculating circle with position vector $\vec{q}$. Thus, in the case of a circle, the Gauss circle can be imaged as osculating circle by rescaling the length of the radius to unity.

3.2 Curves in $\mathbb{R}^3$

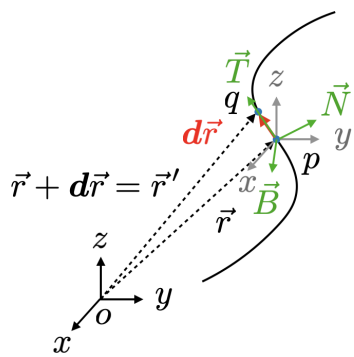

Fig. 3.3: A curve in $\mathbb{R}^3$.

In $\mathbb{R}^3$ we can similarly define the tangent vector $\vec{T}$ and the principal normal vector $\vec{N}$ of a point on a curve. In addition, we can define the so-called *binormal vector* through the *vector product* of $\vec{T}$ and $\vec{N}$

$$\boxed{\vec{B} = \vec{T} \times \vec{N}} \quad \text{or} \quad \boxed{\vec{B} = \vec{T} \wedge \vec{N}.}^{\dagger 4} \tag{3.21}$$

Again, we can compute the derivative of $\vec{N}$ to find the relationship between each component, i.e., the *Frenet-Serret formula* in $\mathbb{R}^3$. By differetiating the orthogonality conditions

$$\begin{cases} \vec{T} \cdot \vec{N} = 0, & (3.22a) \\ \vec{T} \cdot \vec{B} = 0, & (3.22b) \\ \vec{N} \cdot \vec{B} = 0, & (3.22c) \end{cases}$$

we obtain

$$\begin{cases} 0 = \underbrace{\vec{T}\,'}_{\kappa\vec{N}} \cdot \vec{N} + \vec{T} \cdot \vec{N}\,' = \kappa + \vec{T} \cdot \vec{N}\,' & (3.23a) \\ 0 = \underbrace{\vec{T}\,'}_{\kappa\vec{N}} \cdot \vec{B} + \vec{T} \cdot \vec{B}\,' = 0 + \vec{T} \cdot \vec{B}\,', & (3.23b) \\ 0 = \vec{N}\,' \cdot \vec{B} + \vec{N} \cdot \vec{B}\,'. & (3.23c) \end{cases}$$

[†4] The *cross product* $\times$ of two vectors cannot be used in higher dimensional space, generally we can use the *exterior product* $\wedge$ for any dimensions $n \geq 3$.

Because $\{\vec{T}, \vec{N}, \vec{B}\}$ is an orthonormal frame, then we can assume that

$$\vec{N}' = a\vec{T} + b\vec{B} \quad \text{and} \quad \vec{B}' = c\vec{T} + d\vec{N}. \tag{3.24}$$

By the relations (3.23) above, we have

$$a = -\kappa, \quad c = 0 \quad \text{and} \quad d = -b. \tag{3.25}$$

Consequently $\vec{N}'$ and $\vec{B}'$ are respectively written by

$$\vec{N}' = -\kappa\vec{T} + b\vec{B} \quad \text{and} \quad \vec{B}' = -\tau\vec{N}, \tag{3.26}$$

where $\tau(s) \coloneqq b(s)$ is defined as the *torsion* of a curve. Then the Frenet-Serret formula is

$$\begin{cases} \vec{r}' = +\vec{T}, \\ \vec{T}' = +\kappa\vec{N}, \\ \vec{N}' = -\kappa\vec{T} + \tau\vec{B}, \\ \vec{B}' = -\tau\vec{N}, \end{cases} \implies \begin{pmatrix} \vec{r}' \\ \vec{T}' \\ \vec{N}' \\ \vec{B}' \end{pmatrix} = \left(\begin{array}{cc:c} 1 & 0 & 0 \\ 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ \hdashline 0 & -\tau & 0 \end{array}\right) \begin{pmatrix} \vec{T} \\ \vec{N} \\ \vec{B} \end{pmatrix}. \tag{3.27}$$

Appendices

A Curvilinear Coordinates

In many cases it is not convenient to use to Cartesian coordinates for calculations. We review the so-called *curvilinear coordinates* here.

Polar coordinates in $\mathbb{R}^2$

- The displacement vector:

$$\boldsymbol{d}\vec{r} = \vec{e}_r\, dr + \vec{e}_\theta (r\, d\theta). \tag{A.1a}$$

- Vectorial variations:

$$\boldsymbol{d}\vec{e}_r = \vec{e}_\theta, \quad \boldsymbol{d}\vec{e}_\theta = -\,\vec{e}_r. \tag{A.1b}$$

- The line element:

$$dS^2 = dr^2 + r^2\, d\theta^2. \tag{A.1c}$$

- The surface element:

$$\begin{aligned} d\mathcal{A} &= d\ell_{\text{circle}}|_r \cdot dr \\ &= (r\, d\theta) dr = r\, dr\, d\theta. \end{aligned} \tag{A.1d}$$

Cylindrical coordinates in $\mathbb{R}^3$

- The displacement vector:

$$\boldsymbol{d}\vec{r} = \vec{e}_r\, dr + \vec{e}_\varphi (\rho\, d\varphi) + \vec{e}_z\, dz. \tag{A.2a}$$

- Vectorial variations:

$$\boldsymbol{d}\vec{e}_\rho = \vec{e}_\varphi, \quad \boldsymbol{d}\vec{e}_\varphi = -\,\vec{e}_\rho, \quad \boldsymbol{d}\vec{e}_z = 0. \tag{A.2b}$$

- The line element:

$$dS^2 = d\rho^2 + \rho^2\, d\varphi^2 + dz^2. \tag{A.2c}$$

- The surface element vector:

$$\overrightarrow{d\mathcal{A}} = \vec{e}_\rho\, d\mathcal{A}|_\rho + \vec{e}_\varphi\, d\mathcal{A}|_\varphi + \vec{e}_z\, d\mathcal{A}|_z \tag{A.2d}$$

$$= \vec{e}_\rho\, (\rho\, d\varphi) dz + \vec{e}_\varphi\, d\rho\, dz + \vec{e}_z\, d\rho (\rho\, d\varphi). \tag{A.2e}$$

- The volume element:

$$\begin{aligned} d\mathcal{V} &= d\mathcal{A}_{\text{cylinder}}|_\rho \cdot d\rho \\ &= \big[(\rho\ d\varphi) dz\big] d\rho = \rho\, d\rho\, d\varphi\, dz. \end{aligned} \tag{A.2f}$$

Spherical coordinates in $\mathbb{R}^3$

- The displacement vector:

$$\boldsymbol{d}\vec{r} = \vec{e}_r\, dr + \vec{e}_\theta (r\, d\theta) + \vec{e}_\varphi (r \sin\theta\, d\varphi). \tag{A.3a}$$

- Vectorial variations:

$$\begin{aligned} &\boldsymbol{d}\vec{e}_r = \vec{e}_\theta\, d\theta + \vec{e}_\varphi\, (\sin\theta\, d\varphi), \quad \boldsymbol{d}\vec{e}_\theta = -\vec{e}_r\, d\theta, \\ &\boldsymbol{d}\vec{e}_\varphi = -\vec{e}_r \left(\frac{d\varphi}{\sin\theta}\right) + \vec{e}_z (\cot\theta\, d\varphi) = -\vec{e}_r \left(\frac{d\varphi}{\sin\theta}\right) + \frac{1}{2}\left(\vec{e}_r \cos\theta - \vec{e}_\theta \sin\theta\right)(\cot\theta\, d\varphi). \end{aligned} \tag{A.3b}$$

- The line element:

$$dS^2 = dr^2 + r^2\, d\theta^2 + r^2 \sin^2\theta\, d\varphi^2. \tag{A.3c}$$

- The surface element vector:

$$\overrightarrow{d\mathcal{A}} = \vec{e}_r\, d\mathcal{A}|_r + \vec{e}_\theta\, d\mathcal{A}|_\theta + \vec{e}_\varphi\, d\mathcal{A}|_\varphi \tag{A.3d}$$

$$= \vec{e}_r\, (r\, d\theta)(\sin\theta\, d\varphi) + \vec{e}_\theta\, dr (r\, d\theta) + \vec{e}_\varphi\, dr (r \sin\theta\, d\varphi). \tag{A.3e}$$

- The volume element:

$$\begin{aligned} d\mathcal{V} &= d\mathcal{A}_{\text{sphere}}|_r \cdot dr \\ &= \big[(r\, d\theta)(r \sin\theta\, d\varphi)\big] dr = r^2 \sin\theta\, dr\, d\theta\, d\varphi. \end{aligned} \tag{A.3f}$$